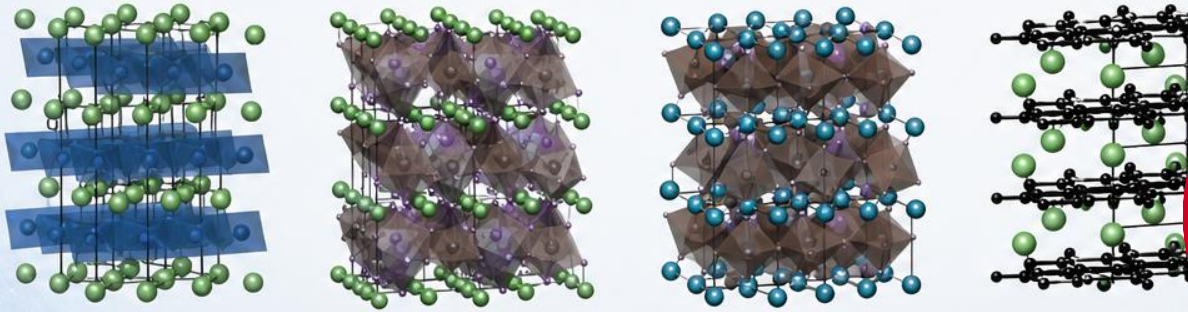
Quantum-Chemical Bonding Analysis of Functional Solid-State Materials

(with Emphasis on Batteries)



Dr. Christina Ertural

Theoretical Chemistry & Materials Informatics

Outline

- Motivation & short introduction to LOBSTER

- Covalency & Ionicity
  - how it can be used to assess the stability of materials and find new suitable candidates

- Alternative battery materials
  - amorphous materials for anodes
  - transition metal free cathodes

Academic background

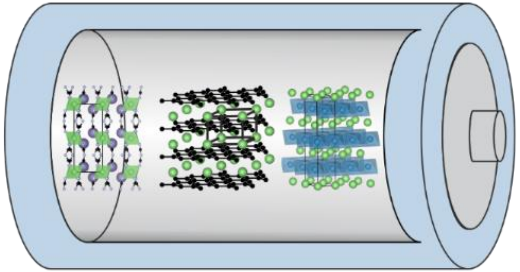
- Study of chemistry (B. Sc. & M. Sc. & PhD) at RWTH Aachen University

- Field of research in Prof. Dronskowski’s group at Inorganic Institute of RWTH Aachen:

– *Theoretical and quantum chemistry*
  > Density functional theory

– *Chemical bonding* in functional materials
  > Solid-state chemistry and materials

– Focus: Method & *C ++ code development* concerning chemical bond descriptors in LOBSTER
  > Mulliken & Löwdin charges, Madelung energy, band-resolved COHP, COBI



LOBSTER
Local-Orbital Basis Suite Towards Electronic-Structure Reconstruction

Nelson, R.; **Ertural, C.**; George, J.; Deringer, V. L.; Hautier, G.; Dronskowski, R., LOBSTER: Local orbital projections, atomic charges, and chemical-bonding analysis from projector-augmented-wave-based density-functional theory. *J. Comput. Chem.* **2020,** *41* (21), 1931–1940.

Academic background

```python
from mp_api.client import MPRester
from autoplex.auto.phonons.flows import CompleteDFTvsMLBenchmarkWorkflow

mpr = MPRester(api_key='YOUR_MP_API_KEY')
structure_list = []
benchmark_structure_list = []
mpids = ["mp-22905"]
# you can put as many mpids as needed e.g. mpids = ["mp-22905", "mp-1185319"] for all LiCl ent
mpbenchmark = ["mp-22905"]
for mpid in mpids:
    structure = mpr.get_structure_by_material_id(mpid)
    structure_list.append(structure)
for mpbm in mpbenchmark:
    bm_structure = mpr.get_structure_by_material_id(mpbm)
    benchmark_structure_list.append(bm_structure)

complete_flow = CompleteDFTvsMLBenchmarkWorkflow(
    apply_data_preprocessing=True,
).make(
    structure_list=structure_list, mp_ids=mpids,
    benchmark_structures=benchmark_structure_list, benchmark_mp_ids=mpbenchmark)

complete_flow.name = "tutorial"
autoplex_flow = complete_flow
```

- Field of research in Prof. George's group at Bundesanstalt für Materialforschung und -prüfung (BAM) Berlin:
  - *Materials informatics*
    > automation, workflows & machine learning
  - *AutoPLEX*
    (automatic potential-landscape explorer)
    > software for generating and benchmarking *machine learning* (ML)-based interatomic potentials
  - Focus: *Python code development* of AutoPLEX from scratch
    > Materials Project framework

Y. Liu, J. D. Morrow, **C. Ertural,** N. L. Fragapane, J. L. A. Gardner, A. A. Naik, Y. Zhou, J. George, V. L. Deringer, Nat Commun 16, 7666 (2025). https://doi.org/10.1038/s41467-025-62510-6.

From theory to chemistry…

Theoretical framework
Schrödinger equation

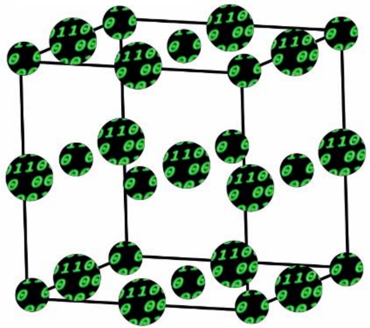
$$\hat{H}\psi = E\psi$$

$$|\psi_j(\mathbf{k})\rangle = \sum_{\mu} C_{\mu j}(\mathbf{k})\,|\chi_\mu(\mathbf{k})\rangle$$

$$\mathrm{COHP}_{\mu\nu} = H_{\mu\nu}\sum_{j,\mathbf{k}} C^*_{\mu j}(\mathbf{k})\ C_{\nu j}(\mathbf{k})$$

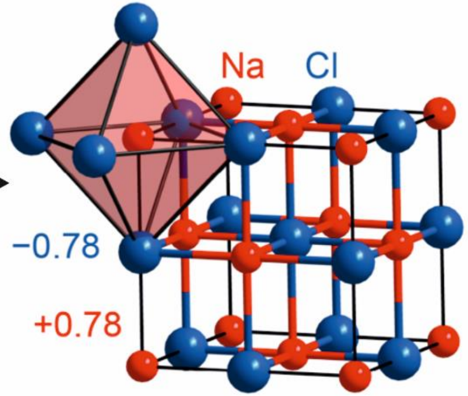
$$q_A = N - \sum_{\mu\in A}\mathrm{GP}_\mu$$

etc.



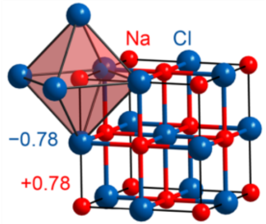
Computer simulation
based on experimental
structure models

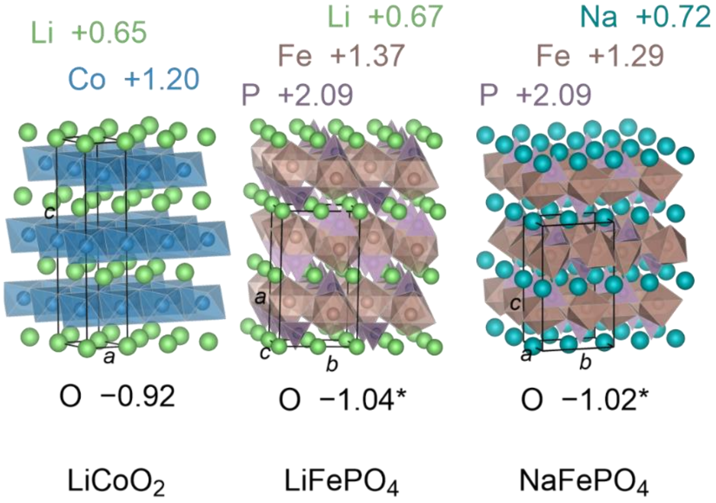
Chemical insights
Structure, charges,
covalency, stability…



…to (everyday) materials

```
MULLIKEN and LOEWDIN POPULATION ANALYSIS

  atom#     atom        Mulliken charge          Loewdin charge
     1        Na                   0.78                    0.67
     5        Cl                  -0.78                   -0.67
          total                   0.00                    0.00


 Mulliken and Loewdin gross orbital population

  atom#     atom    basisfunction         Mulliken GP       Loewdin GP
     1        Na               3s                0.22             0.33
                             2p_y                2.00             2.00
                             2p_z                2.00             2.00
                             2p_x                2.00             2.00
                            total                6.22             6.33

     5        Cl               3s                1.95             1.87
                             3p_y                1.95             1.93
                             3p_z                1.95             1.93
                             3p_x                1.95             1.93
                            total                7.78             7.67
```





Chemical insights

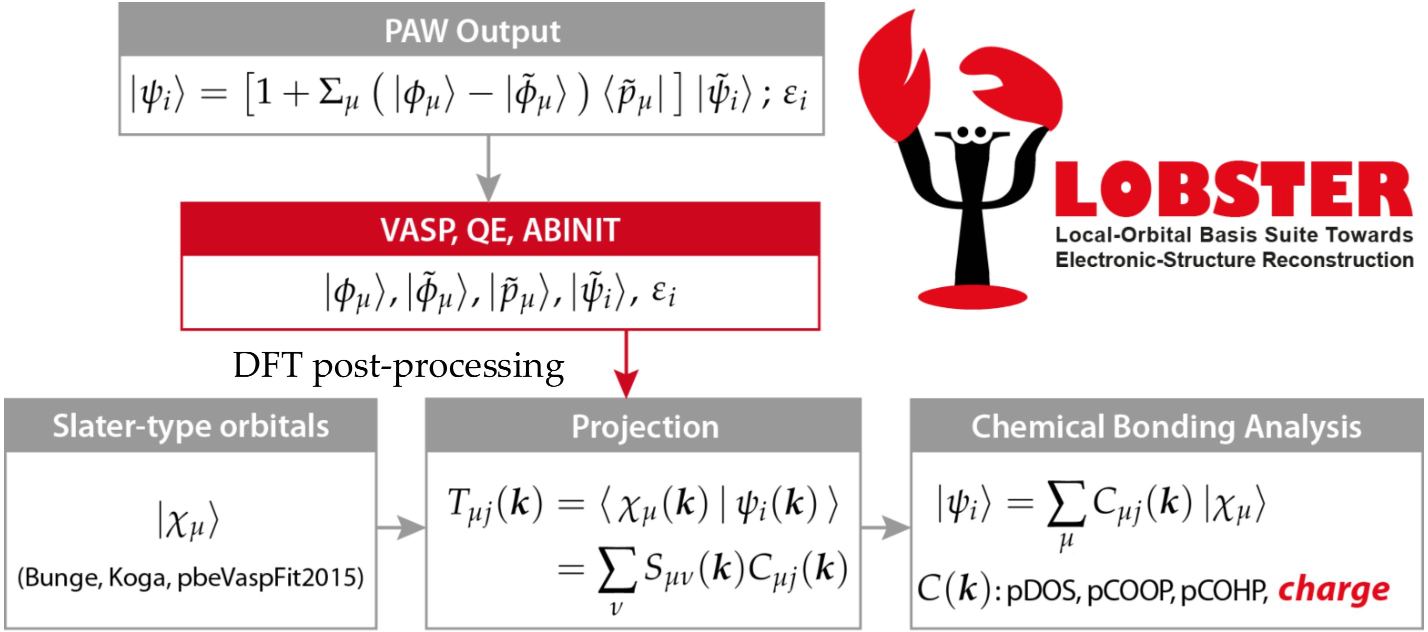
Structure, charges, covalency, stability…

Materials properties

Conductivity, hardness, (dis)charge behaviour…

Ertural, C.; Stoffel, R. P..; Müller, P. C..; Vogt, C. A..; Dronskowski, R., First-Principles Plane-Wave-Based Exploration of Cathode and Anode Materials for Li and Na-ion Batteries involving Complex Nitrogen-based Anions. *Preprint*, **2021**, DOI: 10.33774/chemrxiv-2021-6cxlq.

The right tool: LOBSTER



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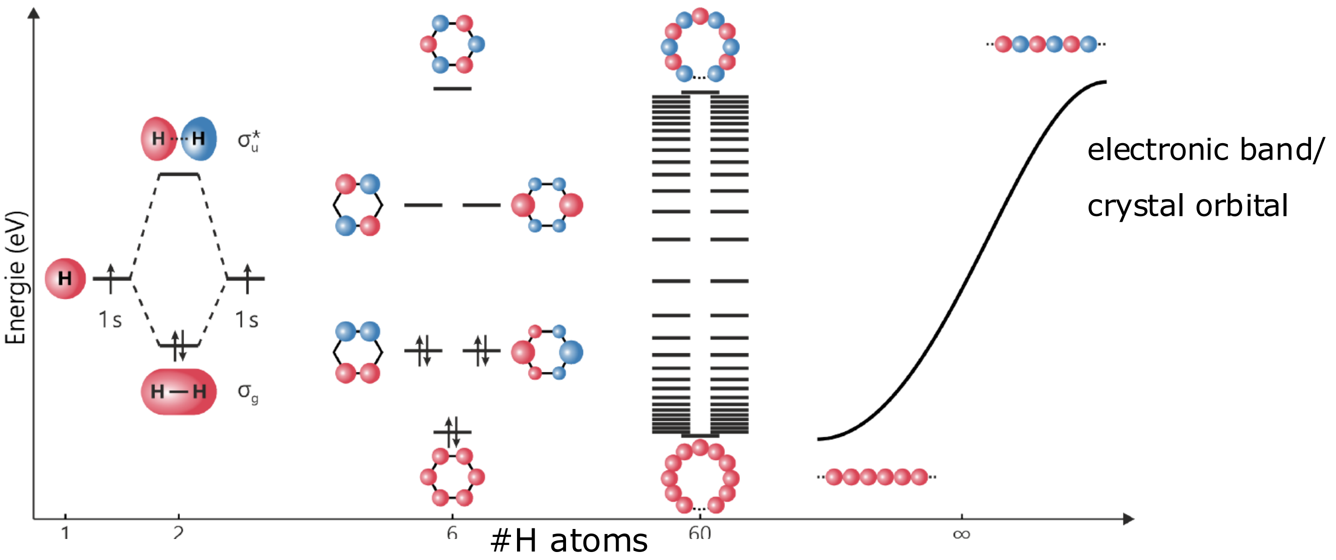
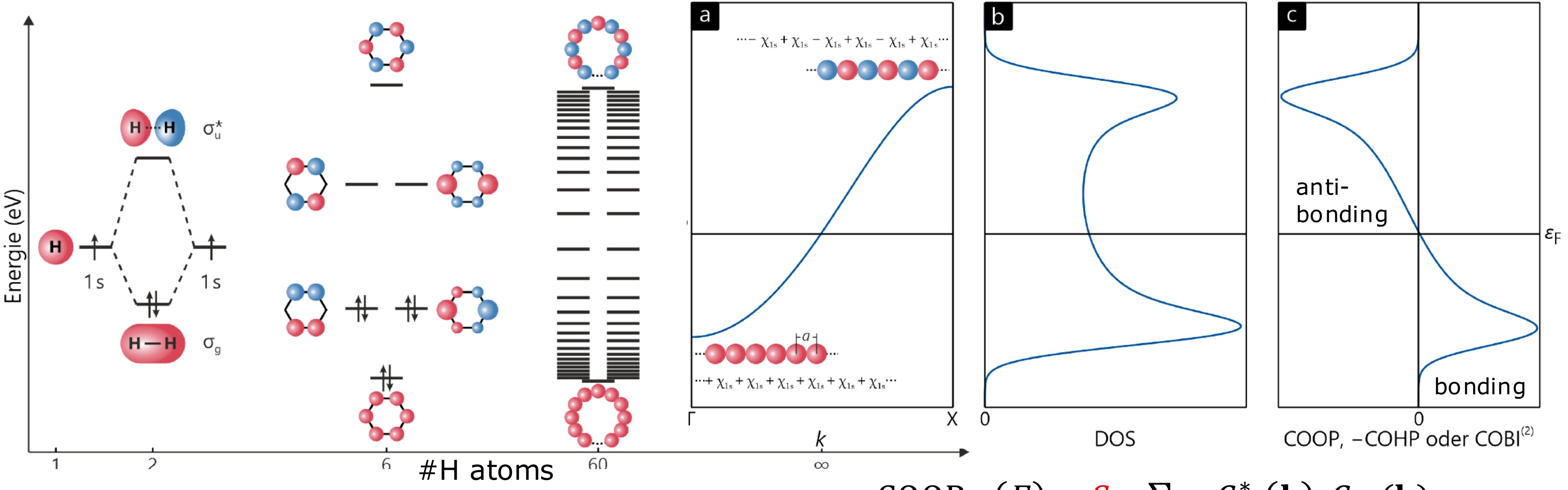
Covalent bond analysis: COOP, COHP



from atoms to molecules to solids

Covalent bond analysis: COOP, COHP

Crystal Orbital Overlap Population & Crystal Orbital Hamilton Population



from atoms to molecules to solids

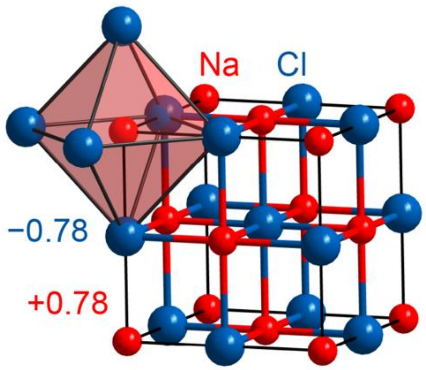
$$\mathrm{COOP}_{\mu\nu}(E) \sim S_{\mu\nu} \sum_{j,\mathbf{k}} C^*_{\mu j}(\mathbf{k})\ C_{\nu j}(\mathbf{k})$$

$$\mathrm{COHP}_{\mu\nu}(E) \sim H_{\mu\nu} \sum_{j,\mathbf{k}} C^*_{\mu j}(\mathbf{k})\ C_{\nu j}(\mathbf{k})$$

Ionic bond analysis

Mulliken & Löwdin population analysis
- wave function based
- basis set dependent (molecular case)

Popular alternative:
Bader charge analysis
- electron density based



LCAO ansatz (**k**-space):

$$|\psi_j(\boldsymbol{k})\rangle = \sum_{\mu} C_{\mu j}(\boldsymbol{k})|\chi_\mu(\boldsymbol{k})\rangle$$

Gross (orbital) population:

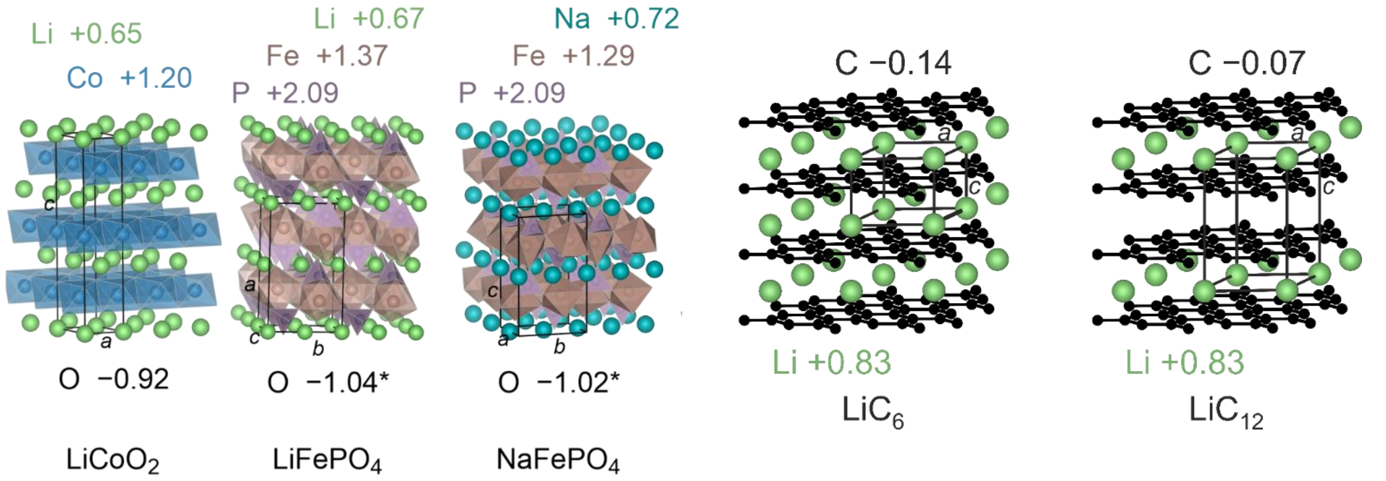
$$\mathrm{GP}_\mu = \sum_{\boldsymbol{k}} \sum_{\nu} P_{\mu\nu}(\boldsymbol{k})\, S_{\mu\nu}(\boldsymbol{k})\; w(\boldsymbol{k})$$

Mulliken/Löwdin charge:

$$q_A = N - \sum_{\mu \in A} \mathrm{GP}_\mu$$

Mulliken & Löwdin charges

- commercially used materials

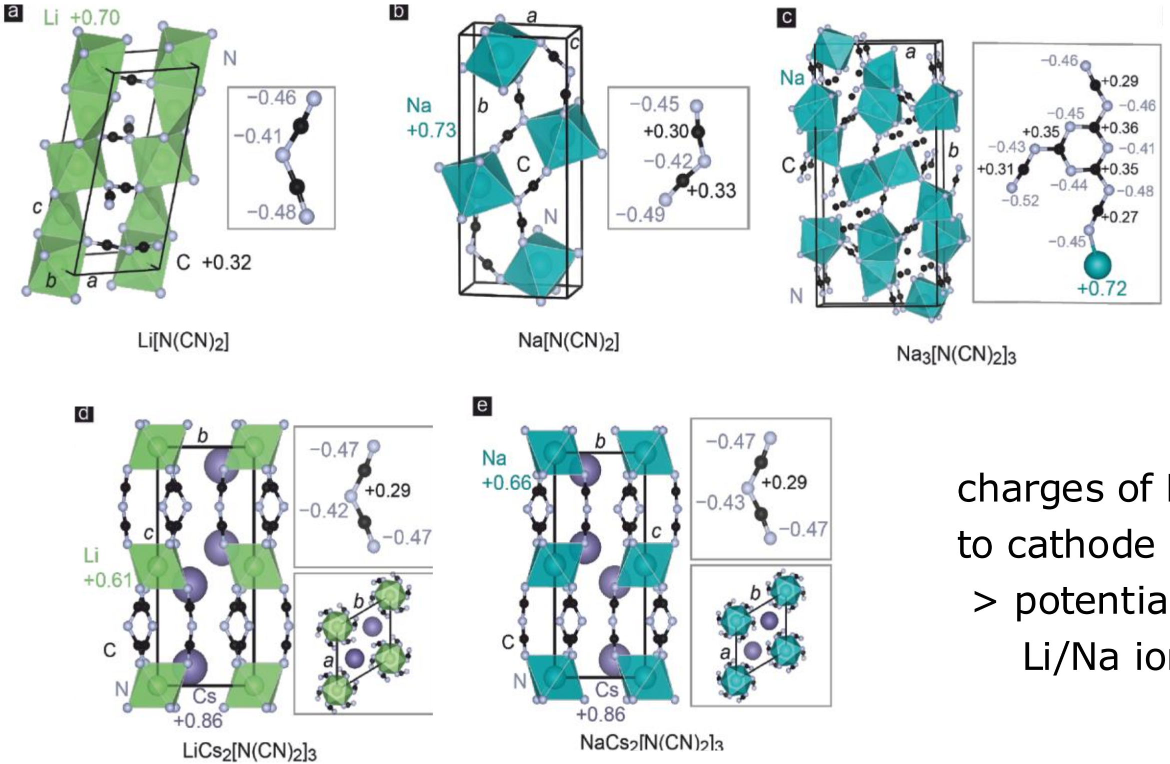


cathodes

anodes

Ertural, C.; Stoffel, R. P..; Müller, P. C..; Vogt, C. A..; Dronskowski, R., First-Principles Plane-Wave-Based Exploration of Cathode and Anode Materials for Li and Na-ion Batteries involving Complex Nitrogen-based Anions. *Chem. Mater.* **2022**, *34*, 652–668.

Mulliken & Löwdin charges: Li/Na dicyanamide salts



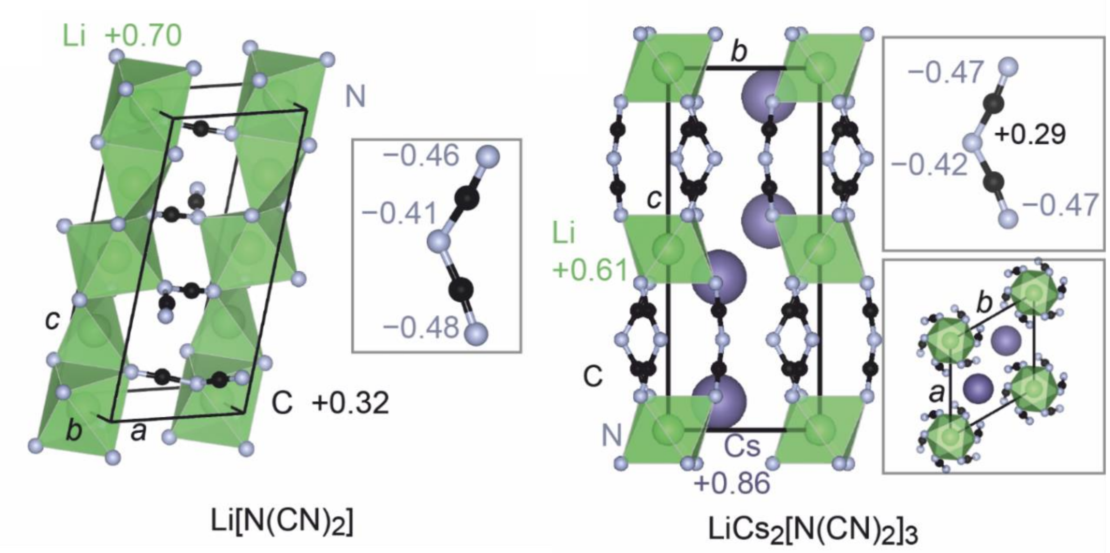
charges of Li and Na comparable to cathode materials

> potential candidates for Li/Na ion battery cathodes

Ertural, C.; Stoffel, R. P..; Müller, P. C..; Vogt, C. A..; Dronskowski, R., First-Principles Plane-Wave-Based Exploration of Cathode and Anode Materials for Li and Na-ion Batteries involving Complex Nitrogen-based Anions. *Chem. Mater.* **2022**, *34*, 652–668.

Mulliken & Löwdin charges

- new transition metal free candidates



$([N(CN)_2])_{2(s)} + 2\,LiC_{6(s)} \rightarrow 2\,Li[N(CN)_2]_{(s)} + 2\,C_{6(s)},$ $\Delta H_r = -390.8\ \text{kJ mol}^{-1} \triangleq$ **+4.1 V**

$Cs_2[N(CN)_2]_{3(s)} + LiC_{6(s)} \rightarrow LiCs_2[N(CN)_2]_{3(s)} + C_{6(s)},$ $\Delta H_r = -444.4\ \text{kJ mol}^{-1} \triangleq$ **+4.6 V**

Ertural, C.; Stoffel, R. P.; Müller, P. C.; Vogt, C. A.; Dronskowski, R., First-Principles Plane-Wave-Based Exploration of Cathode and Anode Materials for Li and Na-ion Batteries involving Complex Nitrogen-based Anions. *Chem. Mater.* **2022**, *34*, 652–668.

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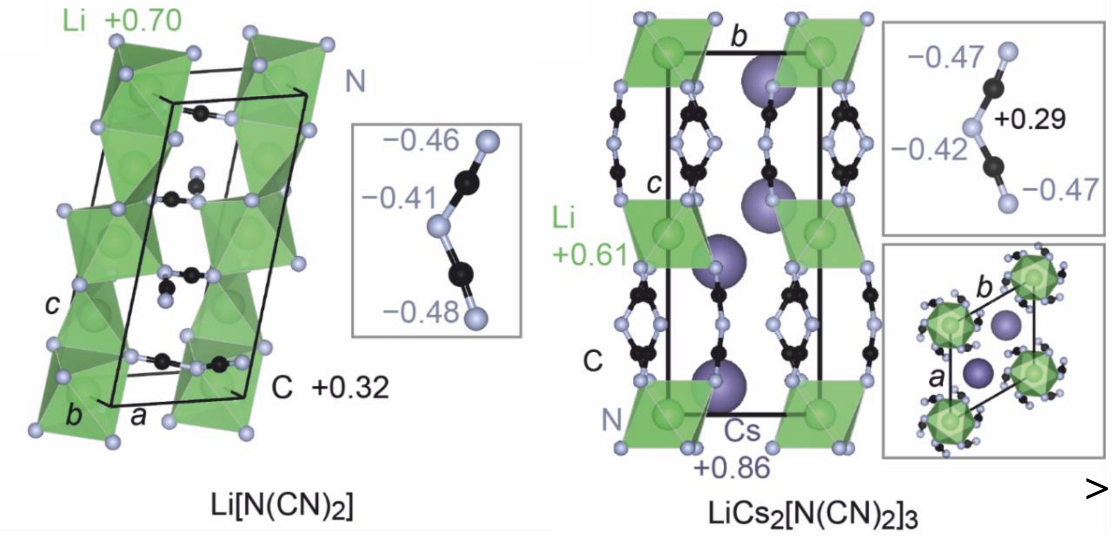

> experimental validation ongoing

$$([N(CN)_2])_{2(s)} + 2\,LiC_{6(s)} \rightarrow 2\,Li[N(CN)_2]_{(s)} + 2\,C_{6(s)}, \qquad \Delta H_r = -390.8\ \mathrm{kJ\ mol^{-1}} \triangleq +4.1\ \mathrm{V}$$

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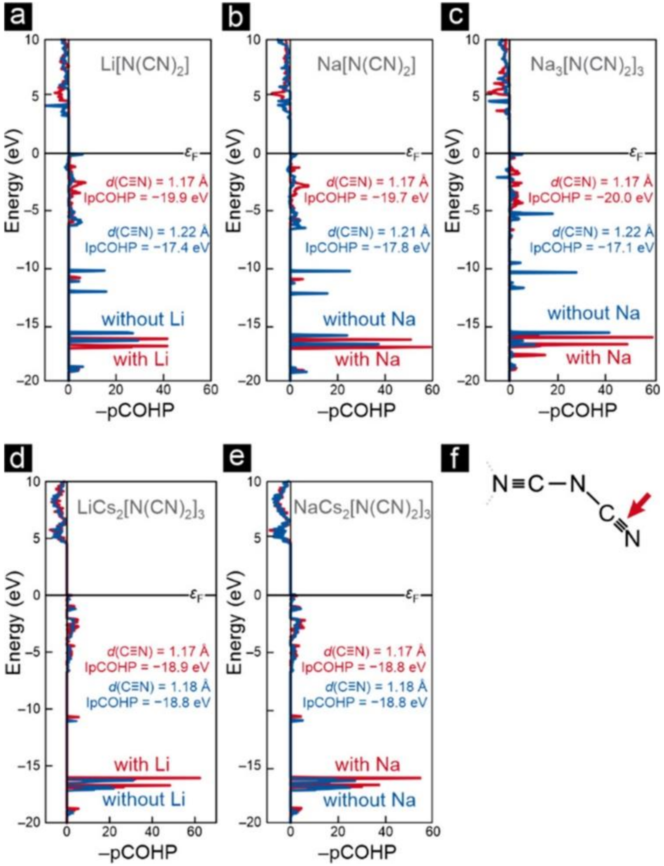
COHP: Deintercalation of Li/Na dicyanamide salts

a lot of change >

almost no change >



COHP (as measure of chemical bond covalency) plot gives hint on structural deformation and change of chemical environment during deintercalation

> using COHP to assess stability

Ertural, C.; Stoffel, R. P..; Müller, P. C..; Vogt, C. A..; Dronskowski, R., First-Principles Plane-Wave-Based Exploration of Cathode and Anode Materials for Li and Na-ion Batteries involving Complex Nitrogen-based Anions. *Chem. Mater.* **2022**, *34*, 652–668.

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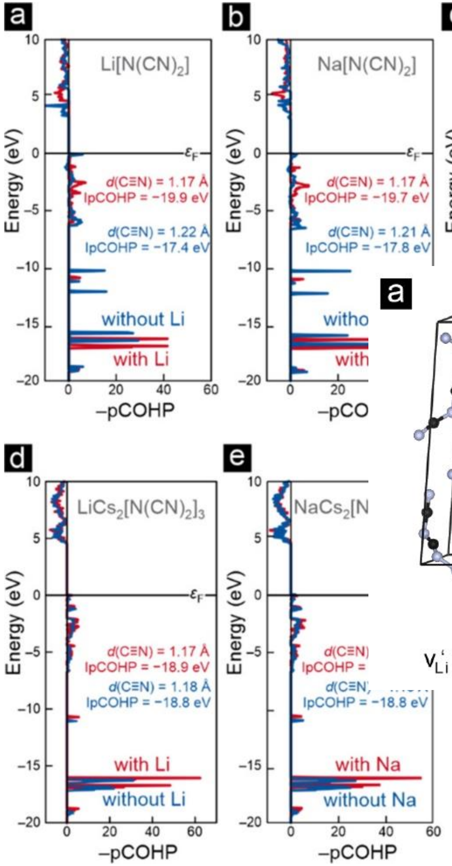
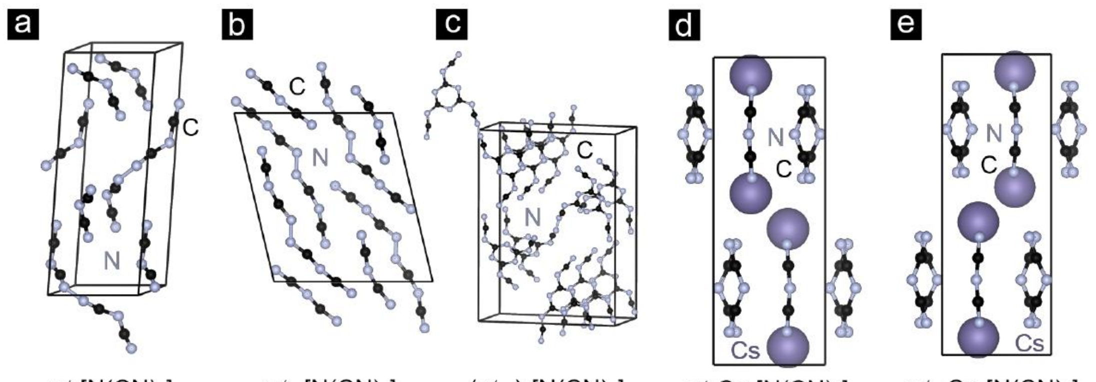


COHP (as measure of chemical bond covalency) plot gives hint on structural deformation and



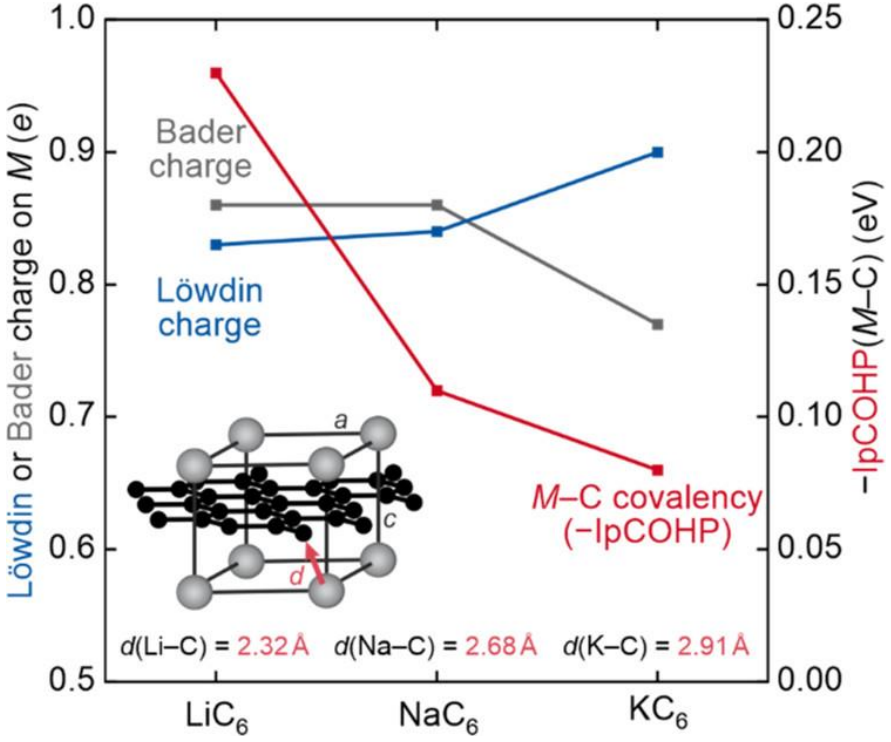
$v'_{Li}[N(CN)_2]$ $v'_{Na}[N(CN)_2]$ $(v'_{Na})_3[N(CN)_2]_3$ $v'_{Li}Cs_2[N(CN)_2]_3$ $v'_{Na}Cs_2[N(CN)_2]_3$

relaxed deintercalated structures

Ertural, C.; Stoffel, R. P..; Müller, P. C..; Vogt, C. A..; Dronskowski, R., First-Principles Plane-Wave-Based Exploration of Cathode and Anode Materials for Li and Na-ion Batteries involving Complex Nitrogen-based Anions. *Chem. Mater.* **2022**, *34*, 652–668.

Charges combined with COHP

- The intercalation problem of Na in (pristine) graphite



> only Li is known to intercalate into pristine graphite, yielding LiC_6

> Na does not intercalate at all

> K favors the composition KC_8 (but stability issues known)

> **why?**

Literature: LiC_6 is stabilized by an additional covalent contribution in *M*–C binding energy

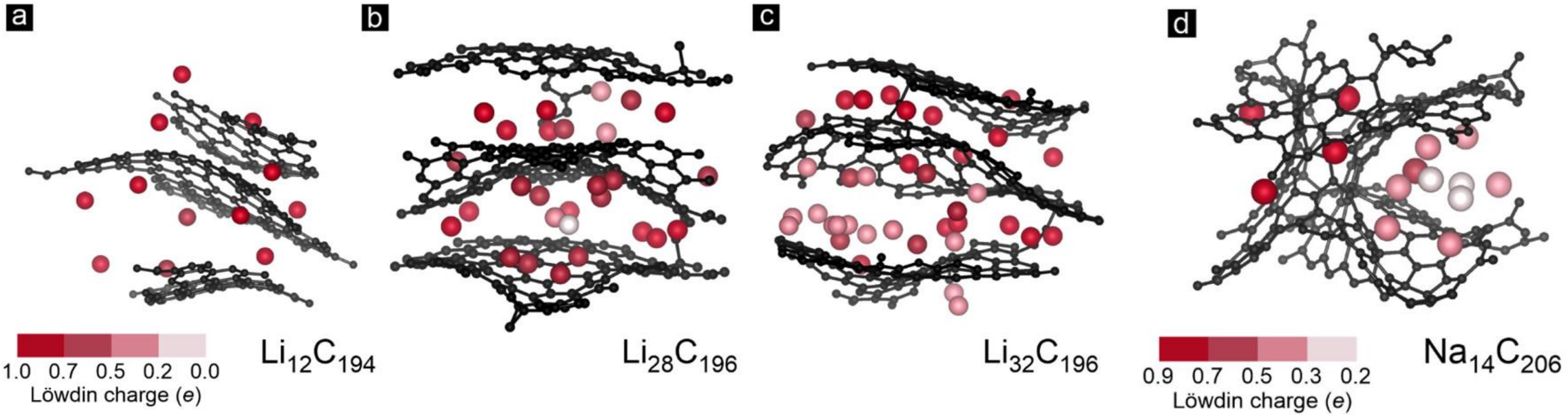
> confirmed by present study

Ertural, C.; Stoffel, R. P.; Müller, P. C.; Vogt, C. A..; Dronskowski, R., First-Principles Plane-Wave-Based Exploration of Cathode and Anode Materials for Li and Na-ion Batteries involving Complex Nitrogen-based Anions. *Chem. Mater.* **2022**, *34*, 652–668.

Lenchuk, O.; Adelhelm, P.; Mollenhauer, D. New Insights Into the Origin of Unstable Sodium Graphite Intercalation Compounds. *Phys. Chem. Chem. Phys.* **2019**, *21*, 19378–19390.

Mulliken & Löwdin charges

- Amorphous carbonaceous materials as alternative to Li/graphite anodes and to solve the intercalation problem of Na in (pristine) graphite

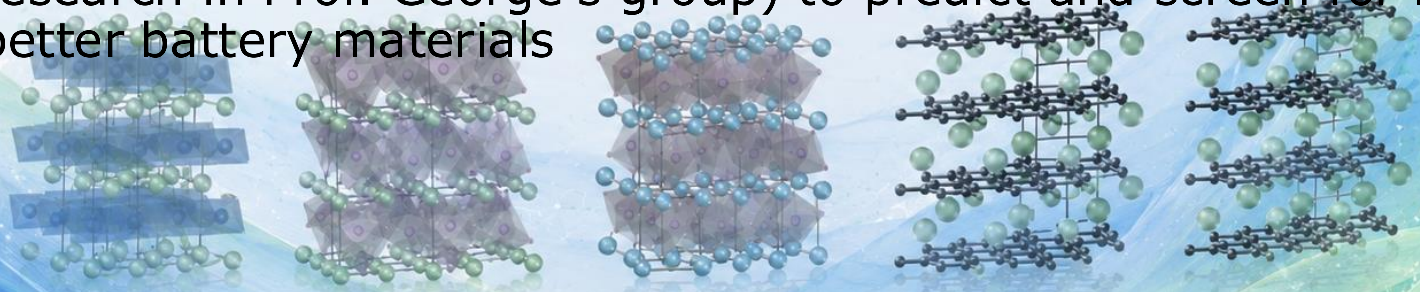


> good Coulombic (charging/discharging) behaviour expected estimated from the Löwdin charges

Ertural, C.; Stoffel, R. P.; Müller, P. C.; Vogt, C. A..; Dronskowski, R., First-Principles Plane-Wave-Based Exploration of Cathode and Anode Materials for Li and Na-ion Batteries involving Complex Nitrogen-based Anions. *Chem. Mater.* **2022**, *34*, 652–668.

Summary

- Commercial battery materials were examined using chemical bonding indicators
- Insights were used to look for alternative materials
  - transition metal free cathode materials
  - graphite-like amorphous anode materials to intercalate Na
- **Future steps:** Use automation workflows and quantum chemical bonding descriptors to train machine learning models (ongoing research in Prof. George's group) to predict and screen for new and better battery materials



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Thank you for your kind attention!

Check out quantumchemist.de to learn more about my background!